2026

HVAC AIR DISTRIBUTION SYSTEM DESIGN

CONCEPTUAL MECHANICAL DESIGN FOR A 3-STOREY OFFICE BUILDING

PREPARED BY:
Arnold Okema

Software Used:

- AutoCAD
- Microsoft Excel
- OpenStudio *(For EnergyPlus load calculations)*

1. Project Brief

1.1 Introduction

This report presents the mechanical design for a conceptual three-storey office building located in London, United Kingdom. The project focuses on the design of the building's HVAC air distribution system, including thermal load analysis, ventilation strategy development, duct sizing, pressure drop analysis, and equipment selection.

The objective of the project was to develop a coordinated and technically justified HVAC design suitable for a modern commercial office environment using established industry design methodologies.

The final system design achieves the following overall performance:

- **Total supply airflow:** 5.02 m^3/s
- **Total heating capacity:** 77.3 kW
- **Total cooling capacity:** 62 kW

1.2 Project Scope

The scope of work undertaken within this project includes:

- Heating and cooling load calculations
- Ventilation rate determination
- Development of the air distribution strategy
- Duct sizing using the **equal friction method**
- Pressure loss analysis including duct fittings and terminal devices
- Plant equipment sizing and selection
- Production of system layout drawings

Cost estimation and economic analysis of the system are outside the scope of this study.

2. Building Description

2.1 General Information

- **Building type:** Commercial office
- **Number of floors:** 3
- **Total floor area:** 1048 m^2

- **Typical floor-to-floor height:** 4 m

Refer to appendix A (A101 – A204) for detailed building dimensions.

Outdoor Design Conditions

- **Winter design temperature:** −4.6°C
- **Summer design temperature:** 27.2°C

These values represent the peak outdoor conditions used for HVAC system design.

3. Design Criteria and Assumptions

3.1 Load Calculation Methodology

Thermal modelling of the building was performed using **OpenStudio**, with simulations executed through the **EnergyPlus** physics engine.

EnergyPlus determines heating and cooling loads using the **heat balance method**, which solves the energy interactions between building surfaces, internal gains, and air systems within each thermal zone.

The simulation model accounted for:

- Variable thermostat setpoints across different thermal zones
- Occupancy schedules based on space function
- Lighting and equipment load diversity profiles
- Solar gains determined by building orientation and glazing ratios
- Infiltration and mechanical ventilation rates

This dynamic modelling approach allows peak loads to be determined under realistic operating conditions rather than relying on simplified steady-state assumptions.

3.2 Internal Environmental Criteria

Thermostat setpoints were assigned based on the functional requirements of each space type.

Temperature Setpoints

Open-plan office areas

- Cooling setpoint: 24°C
- Heating setpoint: 21°C

Closed offices and meeting rooms

- Cooling setpoint: 22°C
- Heating setpoint: 20°C

Lobby

- Cooling setpoint: 26°C
- Heating setpoint: 18°C

Occupancy Density

Peak occupancy densities were defined according to space type:

- **Open offices:** 0.1 people/m^2
- **Closed offices:** 0.05 people/m^2
- **Meeting rooms:** 0.5 people/m^2
- **Lobby:** 0.2 people/m^2

Lighting and equipment loads were applied using diversified operating schedules to reflect realistic building operation rather than assuming coincident peak loads.

3.3 Weather Data and Peak Conditions

The simulation utilised a location-specific weather file:

London/Gatwick – GBR IWEC Data (WMO# = 03776)

This dataset represents a **Typical Meteorological Year (TMY)** for the London region.

Peak loads were extracted under the following outdoor conditions:

- **Maximum summer dry-bulb temperature:** 27.2°C
- **Minimum winter dry-bulb temperature:** −4.6°C

These temperatures correspond to peak hourly values identified within the annual simulation dataset.

4. Thermal Load Results

4.1 Dynamic Peak Extraction

Peak heating and cooling loads were determined from hourly simulation outputs.

The model incorporated the following heat transfer mechanisms:

- Transmission heat gains and losses through walls, glazing, roof, and ground slab
- Solar radiation gains through glazed façades
- Internal heat gains from occupants, lighting, and equipment
- Ventilation heat gains and losses due to outdoor air supply

4.2 Load Summary (Peak Hour)

Floor	Peak Cooling Load (kW)	Peak Heating Load (kW)
Ground Floor	21.3	24.7
First Floor	20.3	26.1
Second Floor	20.4	26.5
Total	**62**	**77.3**

These values represent the maximum instantaneous loads extracted from the dynamic simulation results.

5. Air Distribution Strategy

5.1 System Overview

The building is served by a **central Rooftop Unit (RTU)** which provides conditioned supply air to all occupied and conditioned spaces.

The HVAC system consists of:

- Central rooftop unit (RTU)
- Supply ductwork distribution network
- Return air system
- Vertical duct risers serving each floor
- Dedicated mechanical toilet extract system

The **total supply airflow for the building is 5.02 m^3/s.**

5.2 Ductwork Configuration

To minimise friction losses and improve airflow efficiency, **circular ductwork** was adopted for the distribution system.

Compared with rectangular ducts, circular ducts provide:

- Lower pressure losses due to reduced perimeter
- Improved airflow characteristics
- Better structural efficiency

Duct sizes were selected to balance acceptable air velocities with manageable pressure drops, ensuring efficient fan operation and good overall energy performance.

Refer to Appendix A (M101 – M203) for detailed layout drawings.

5.3 Return Air Strategy

A **ceiling plenum return system** was selected to reduce system cost and simplify duct routing.

Return air is drawn from conditioned spaces through ceiling-mounted return grilles into the ceiling void, which acts as a return air plenum. From the plenum, air is conveyed vertically through a return riser back to the rooftop unit, where it is either recirculated or exhausted.

This approach reduces the amount of return ductwork within occupied spaces while maintaining effective airflow distribution.

5.4 Toilet Extract System

A dedicated mechanical extract system serves all toilet facilities within the building.

The toilet extract system is designed to:

- Maintain toilets under **negative pressure** relative to adjacent occupied spaces
- Prevent odour transfer and cross-contamination
- Meet ventilation requirements for sanitary spaces

Extract air is discharged at roof level through a dedicated extract fan sized to meet the calculated airflow requirements.

6. Duct Sizing and Pressure Drop Analysis

6.1 Methodology

Ductwork was sized using the **Equal Friction Method**, which maintains a consistent friction rate throughout the system to achieve balanced pressure characteristics.

The following design friction rates were adopted:

- **RTU supply and return ducts:** 0.45 Pa/m
- **Toilet extract ducts:** 0.8 Pa/m

6.2 Pressure Loss Calculation

The pressure drop analysis considered all major components of the air distribution system, including:

- Straight duct friction losses
- Elbows (90° radius fittings)
- Tee branch and straight-through losses
- Volume control dampers
- Terminal device resistance

The calculated **critical path pressure losses** from the RTU to each floor are summarised below.

Floor	Pressure Loss (Pa)
Second Floor	95.6
First Floor	97.6
Ground Floor	95.1

The **first-floor path** represents the highest resistance and was therefore used as the basis for supply fan sizing.

Including a **10% design margin** to account for system balancing and installation variability, the required **external static pressure** of the RTU supply fan was determined to be:

110 Pa

For the toilet extract system, the calculated critical path pressure loss was **40 Pa**, which was used as the design external static pressure for the extract fan.

7. Equipment Selection

Based on the calculated airflow and peak load requirements, the following equipment was selected.

Rooftop Unit (RTU)

- **Type:** Packaged rooftop unit
- **Heating coil capacity:** 77.3 kW
- **Cooling coil capacity:** 62 kW
- **Supply fan duty:** 5.02 m^3/s @ 310 Pa
  (Assuming 200 Pa internal static pressure)
- **Estimated fan power:** 2.4 kW
  (Assuming 70% fan efficiency)

Toilet Extract Fan

- **Fan type:** Inline centrifugal fan
- **Fan duty:** 0.09 m^3/s @ 40 Pa
- **Estimated fan power:** 0.0065 kW
  (Assuming 55% fan efficiency)

8. Appendices

Appendix A: Building and HVAC Layout drawings

1 GROUND FLOOR PLAN

Notes

Author

Arnold Okema

Project

Office HVAC

Date

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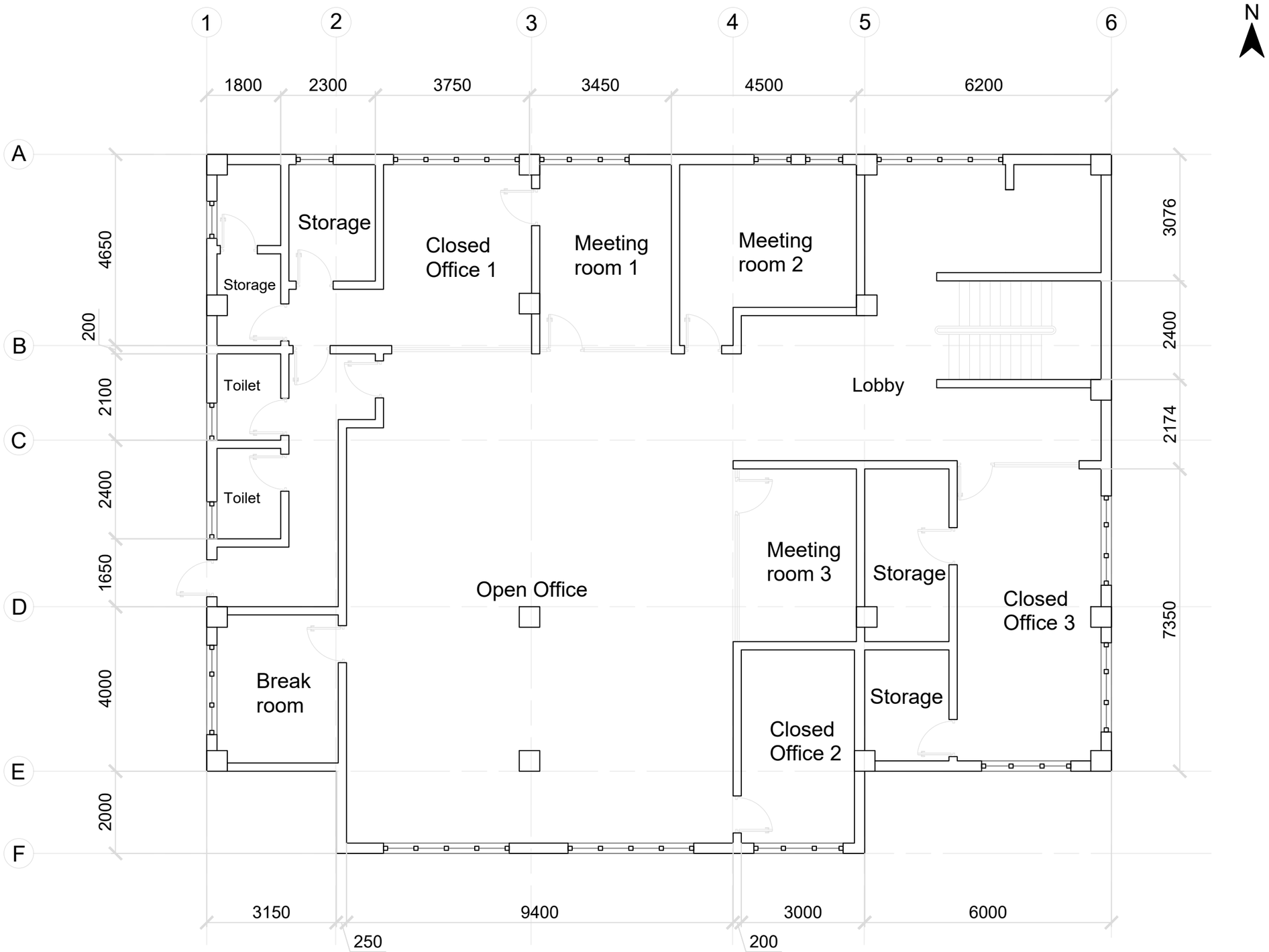

1 FIRST AND SECOND FLOOR PLAN

Notes

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1 SOUTH ELEVATION

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1 WEST ELEVATION

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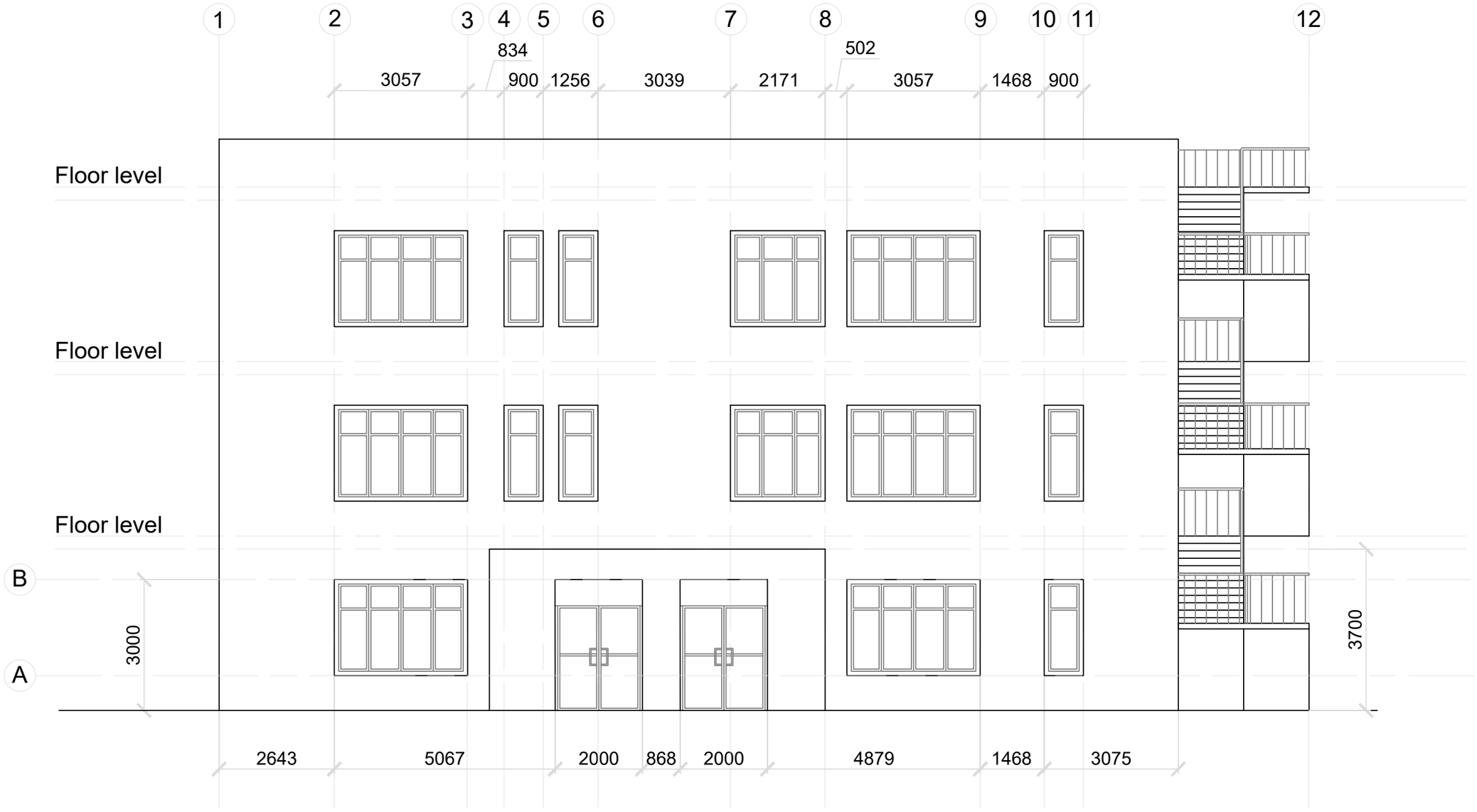

1 NORTH ELEVATION

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1 EAST ELEVATION

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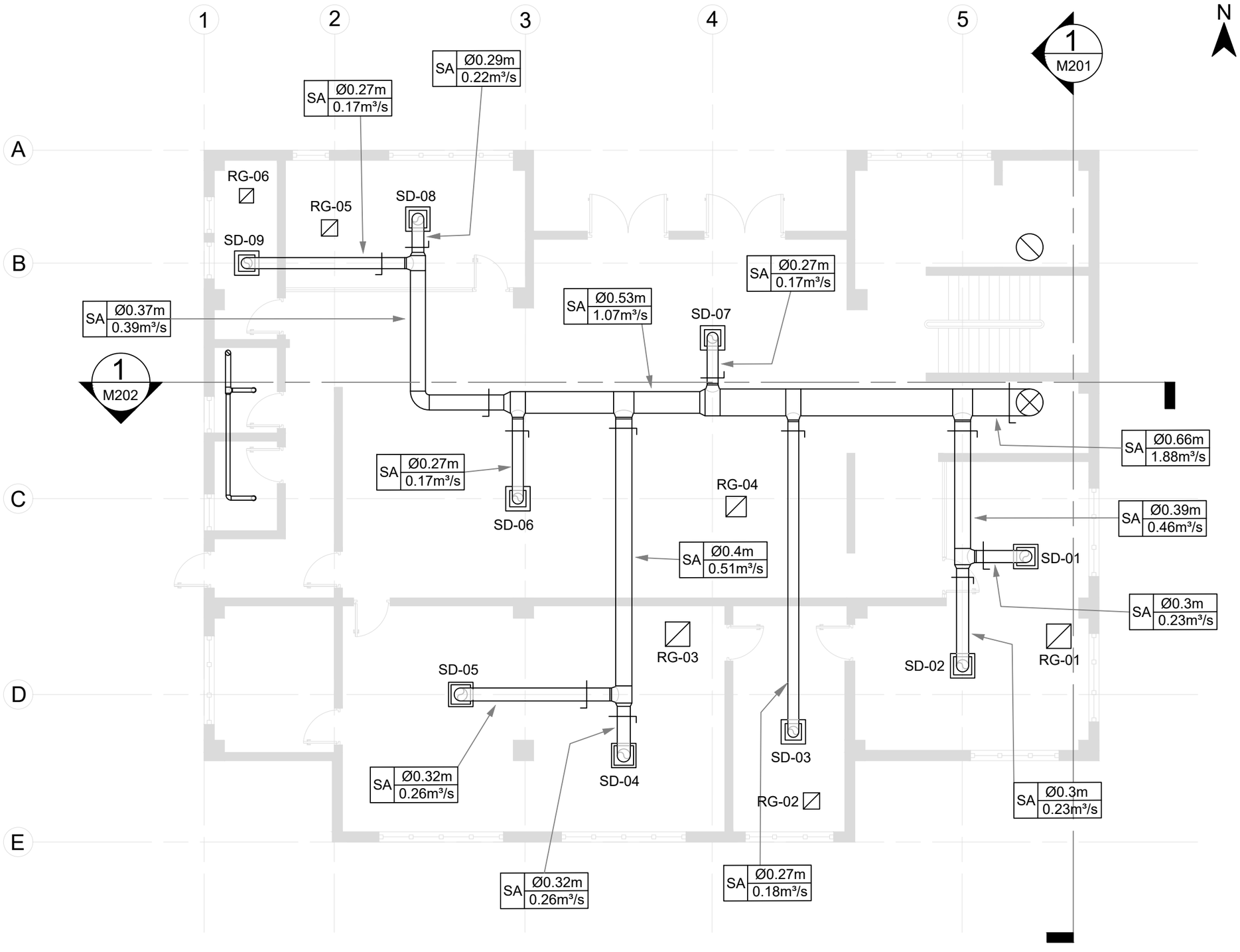

1 GROUND FLOOR HVAC LAYOUT

Notes

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GENERAL NOTES:

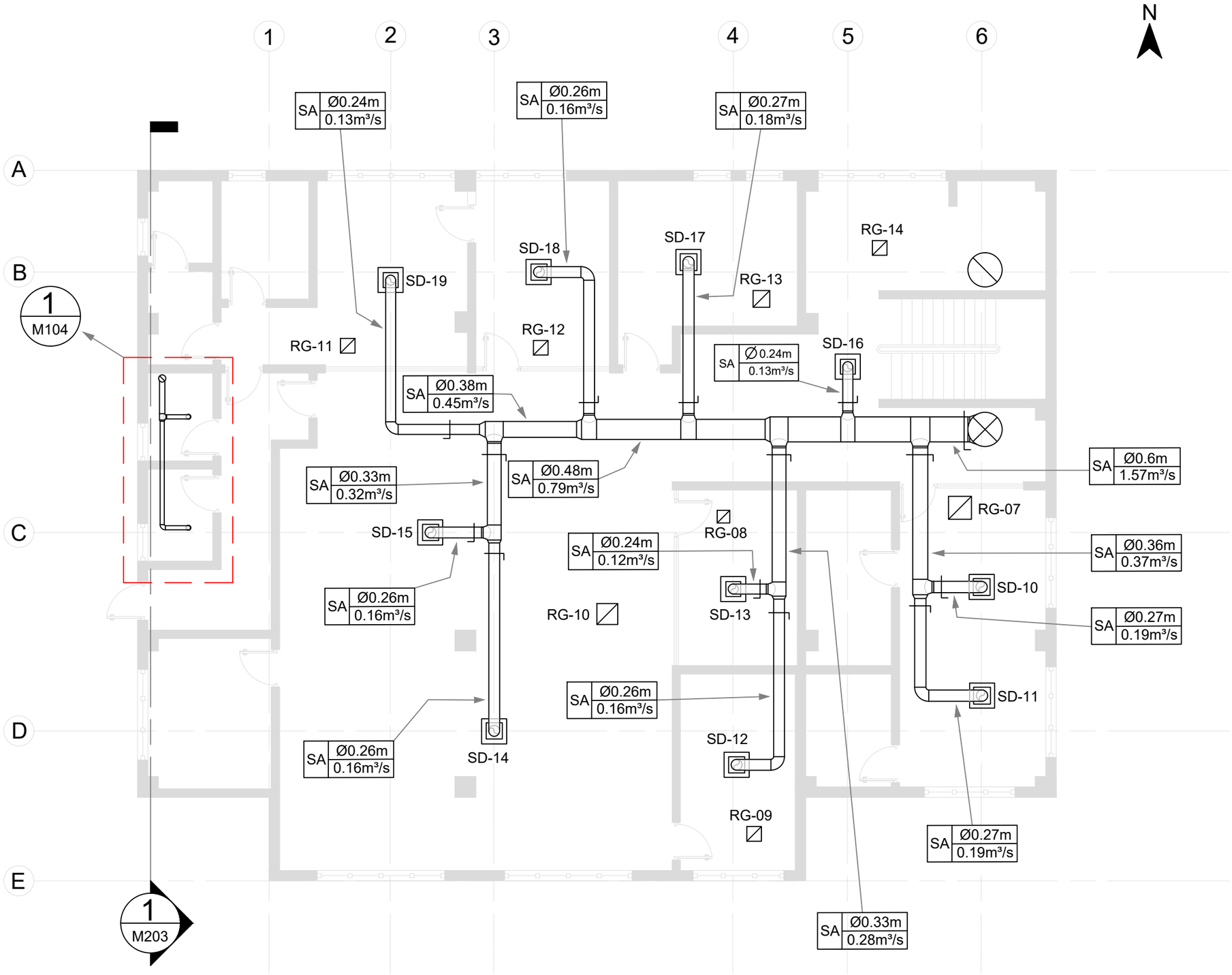
1. Intermittently occupied spaces including toilets, corridors, and break rooms are not provided with dedicated supply air terminals.

2. These spaces are ventilated through transfer air from adjacent conditioned spaces via door undercuts and internal partition openings.

3. To facilitate return air circulation, openings are provided at ceiling plenum level above internal doors and glazed partitions.

4. The ceiling void functions as a return air plenum, allowing return air to flow freely toward the return riser serving the rooftop unit.

5. The arrangement ensures adequate air movement between spaces while maintaining the required negative pressure in toilet areas relative to adjacent occupied zones.



1 FIRST AND SECOND FLOOR HVAC LAYOUT

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GENERAL NOTES:

1. The calculated index run pressure loss is 97.6Pa.

2. 10% was added as a safety margin to account for installation variability.

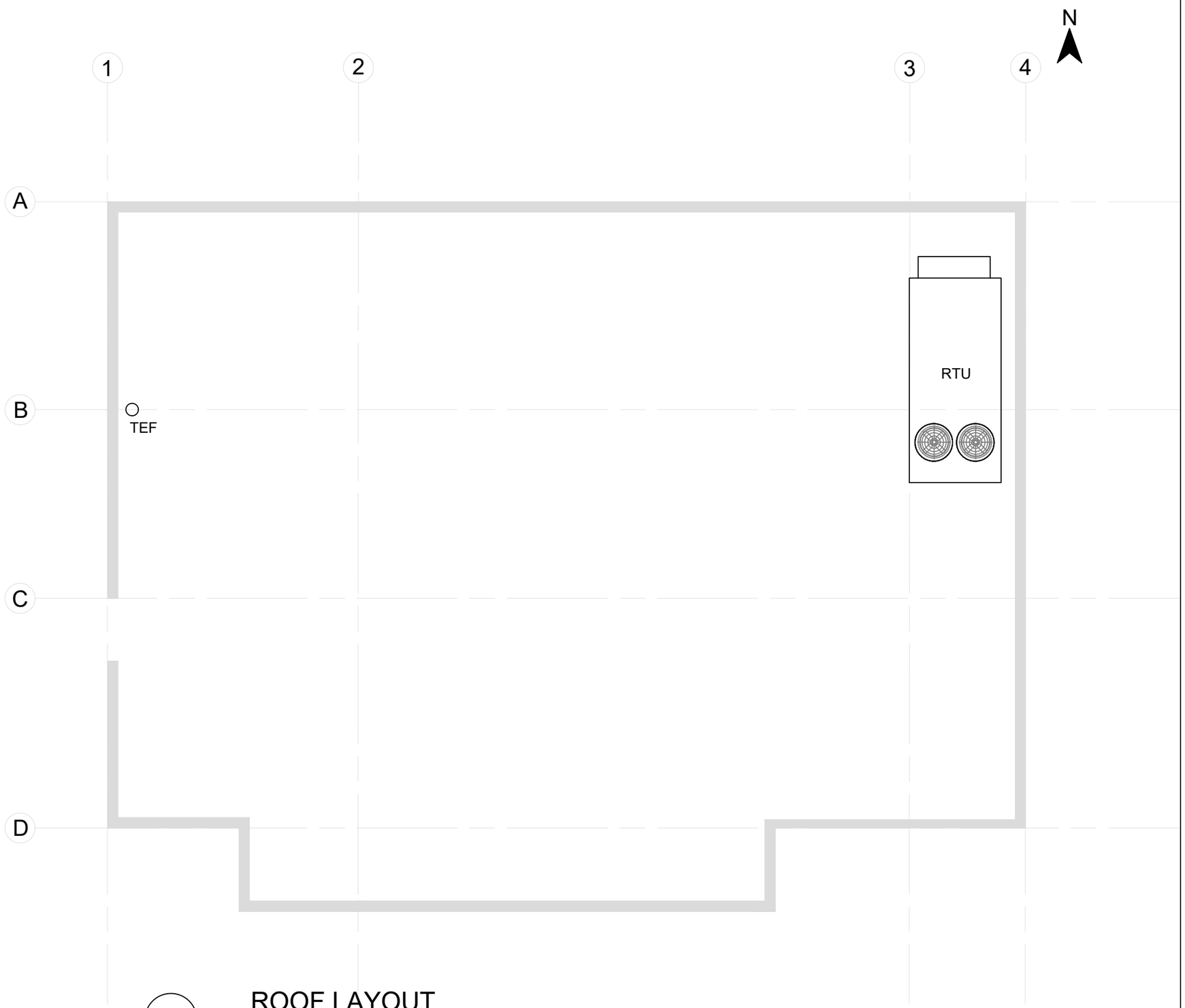
3. Final calculated index run pressure loss is 107.4Pa. Value rounded up to 110Pa.

4. RTU fan External Static Pressure is 110Pa.

5. RTU components assumed to cause internal static pressure of 200Pa.

6. Total RTU fan static pressure is 310Pa.

7. 2.4KW fan power calculated, based on 70% assumed efficiency.



1 ROOF LAYOUT

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GENERAL NOTES:

1. Each toilet is provided with mechanical extract ventilation at a rate of 0.015 m³/s per toilet.

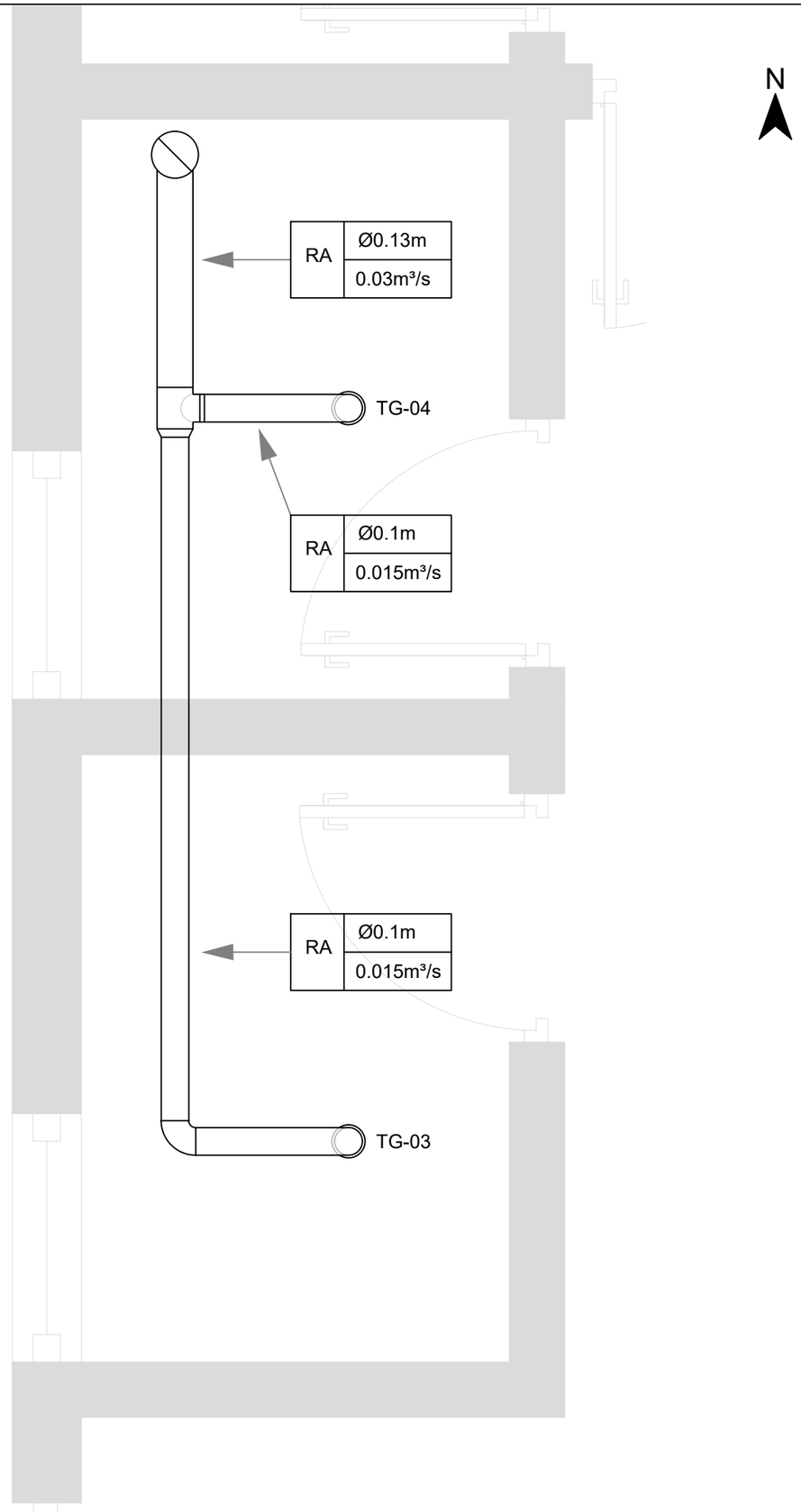
2. The extract system is designed to maintain toilet spaces under negative pressure relative to adjacent occupied areas, preventing odour migration and cross-contamination.

3. Extract air is conveyed through dedicated ductwork to a roof-mounted extract fan.

4. The fan discharges air to atmosphere at roof level.

5. No return or recirculated air from toilet spaces is permitted. All extracted air is discharged directly to atmosphere.

6. Replacement air is provided via transfer air from adjacent spaces through door undercuts.



1 ENLARGED TOILET HVAC LAYOUT

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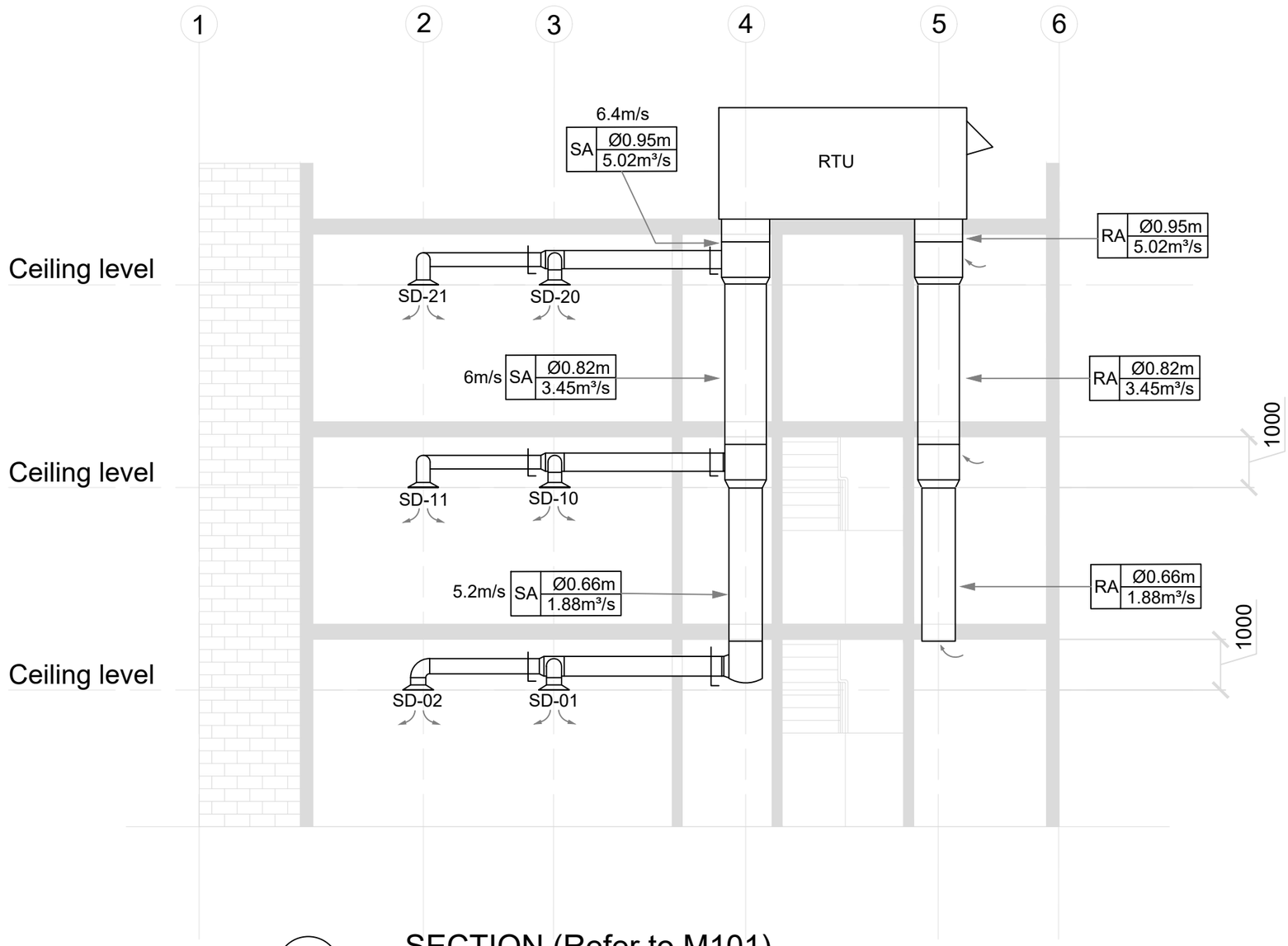
GENERAL NOTES:

1. The ceiling void is used as a plenum to return air to the RTU.

2. The negative pressure created by the RTU fan sucks the return air from the conditioned spaces up into the plenum.

3. The air then flows to the return riser and finds its way back to the RTU where it is either recirculated or expelled into the atmosphere.



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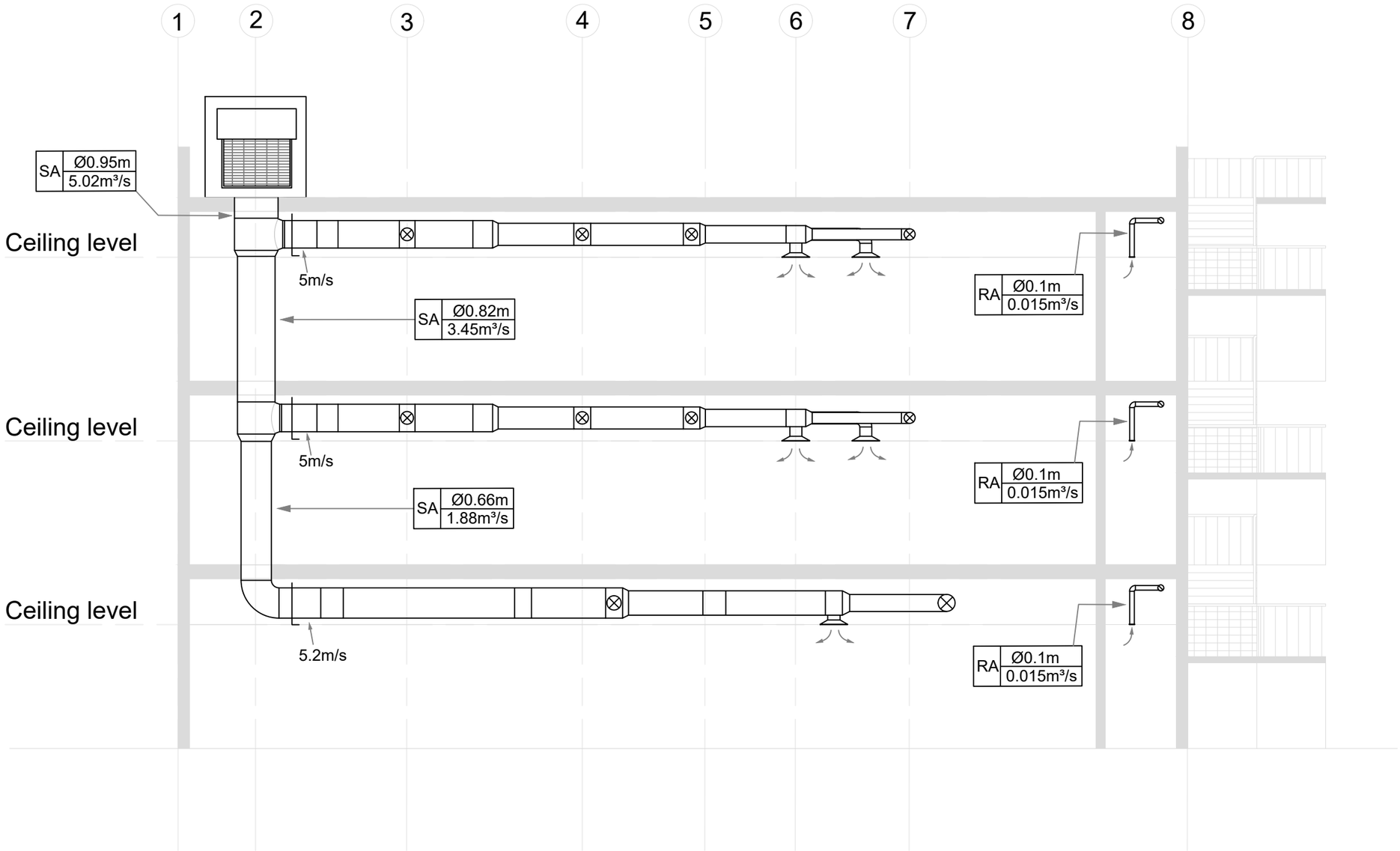

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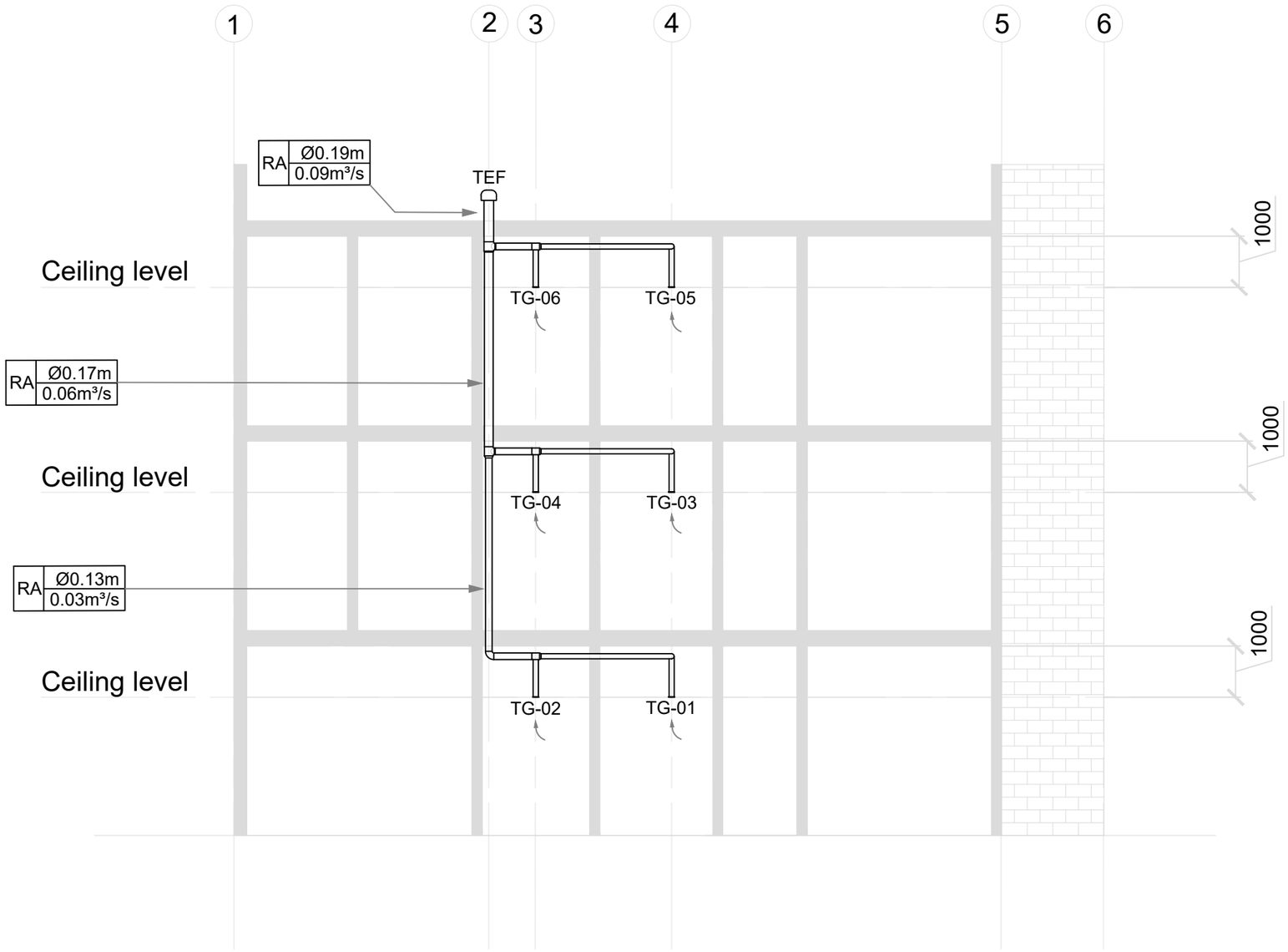

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SUPPLY DIFFUSER SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	NECK SIZE (mm)	AIR FLOW (m^3/s)	AIR VELOCITY (m/s)	FLOOR	LOCATION
SD-01	4-Way ceiling diffuser	600x600	300	0.23	3.2	Ground	Office2
SD-02	4-Way ceiling diffuser	600x600	300	0.23	3.2	Ground	Office2
SD-03	4-Way ceiling diffuser	600x600	270	0.18	2.9	Ground	MeetingRM2
SD-04	4-Way ceiling diffuser	600x600	320	0.26	3.2	Ground	Office1
SD-05	4-Way ceiling diffuser	600x600	320	0.26	3.2	Ground	Office1
SD-06	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lobby
SD-07	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lobby
SD-08	4-Way ceiling diffuser	600x600	290	0.22	3.2	Ground	MeetingRM1
SD-09	4-Way ceiling diffuser	600x600	270	0.17	2.8	Ground	Lounge
SD-10	4-Way ceiling diffuser	600x600	270	0.19	2.9	First	Office 3
SD-11	4-Way ceiling diffuser	600x600	270	0.19	2.9	First	Office 3
SD-12	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	Office 2
SD-13	4-Way ceiling diffuser	600x600	240	0.12	2.7	First	MeetingRM3
SD-14	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	OpenOffice
SD-15	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	OpenOffice
SD-16	4-Way ceiling diffuser	600x600	240	0.13	2.7	First	Lobby
SD-17	4-Way ceiling diffuser	600x600	270	0.18	2.9	First	MeetingRM2
SD-18	4-Way ceiling diffuser	600x600	260	0.16	2.8	First	MeetingRM1
SD-19	4-Way ceiling diffuser	600x600	240	0.13	2.7	First	Office1
SD-20	4-Way ceiling diffuser	600x600	270	0.19	2.9	Second	Office 3
SD-21	4-Way ceiling diffuser	600x600	270	0.19	2.9	Second	Office 3
SD-22	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	Office 2
SD-23	4-Way ceiling diffuser	600x600	240	0.12	2.7	Second	MeetingRM3
SD-24	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	OpenOffice
SD-25	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	OpenOffice
SD-26	4-Way ceiling diffuser	600x600	240	0.13	2.7	Second	Lobby
SD-27	4-Way ceiling diffuser	600x600	270	0.18	2.9	Second	MeetingRM2
SD-28	4-Way ceiling diffuser	600x600	260	0.16	2.8	Second	MeetingRM1
SD-29	4-Way ceiling diffuser	600x600	240	0.13	2.7	Second	Office1

MECHANICAL LEGEND	
SA	Supply air
RA	Return air
SD	Supply diffuser
RG	Return grille
TG	Toilet grille
RTU	Rooftop unit
TEF	Toilet extract fan
	Volume control damper
	Supply duct section
	Return duct section
	Section marker
	Enlarged detail marker

Notes

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RETURN GRILLE SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	AIR FLOW (m³/s)	MOUNTING	FLOOR	LOCATION
RG-01	Eggcrate grille	600x600	0.46	Ceiling	Ground	Office 2
RG-02	Eggcrate grille	400x400	0.18	Ceiling	Ground	MeetingRM2
RG-03	Eggcrate grille	600x600	0.51	Ceiling	Ground	Office 1
RG-04	Eggcrate grille	500x500	0.34	Ceiling	Ground	Lobby
RG-05	Eggcrate grille	400x400	0.22	Ceiling	Ground	MeetingRM1
RG-06	Eggcrate grille	350x350	0.17	Ceiling	Ground	Lounge
RG-07	Eggcrate grille	550x550	0.37	Ceiling	First	Office 3
RG-08	Eggcrate grille	300x300	0.12	Ceiling	First	MeetingRM3
RG-09	Eggcrate grille	350x350	0.16	Ceiling	First	Office 2
RG-10	Eggcrate grille	500x500	0.32	Ceiling	First	OpenOffice
RG-11	Eggcrate grille	350x350	0.13	Ceiling	First	Office 1
RG-12	Eggcrate grille	350x350	0.16	Ceiling	First	MeetingRM1
RG-13	Eggcrate grille	400x400	0.18	Ceiling	First	MeetingRM2
RG-14	Eggcrate grille	350x350	0.13	Ceiling	First	Lobby
RG-15	Eggcrate grille	550x550	0.37	Ceiling	Second	Office 3
RG-16	Eggcrate grille	300x300	0.12	Ceiling	Second	MeetingRM3
RG-17	Eggcrate grille	350x350	0.16	Ceiling	Second	Office 2
RG-18	Eggcrate grille	500x500	0.32	Ceiling	Second	OpenOffice
RG-19	Eggcrate grille	350x350	0.13	Ceiling	Second	Office 1
RG-20	Eggcrate grille	350x350	0.16	Ceiling	Second	MeetingRM1
RG-21	Eggcrate grille	400x400	0.18	Ceiling	Second	MeetingRM2
RG-22	Eggcrate grille	350x350	0.13	Ceiling	Second	Lobby

TOILET GRILLE SCHEDULE

TAG	TYPE	DIMENSIONS (mm)	AIR FLOW (m³/s)	MOUNTING	FLOOR
TG-01	Circular	Ø100	0.015	Ceiling	Ground
TG-02	Circular	Ø100	0.015	Ceiling	Ground
TG-03	Circular	Ø100	0.015	Ceiling	First
TG-04	Circular	Ø100	0.015	Ceiling	First
TG-05	Circular	Ø100	0.015	Ceiling	Second
TG-06	Circular	Ø100	0.015	Ceiling	Second

RTU SCHEDULE

TAG	UNIT TYPE	SUPPLY AIR FLOW (m³/s)	EXTERNAL STATIC PRESSURE (Pa)	COOLING CAPACITY (KW)	HEATING CAPACITY (KW)	FAN POWER (KW)	LOCATION
RTU	Packaged Rooftop	5.02	110	62	77.3	2.4	Roof

TOILET FAN SCHEDULE

TAG	FAN TYPE	AIR FLOW (m³/s)	EXTERNAL STATIC PRESSURE (Pa)	FAN POWER (KW)	LOCATION
TEF	Centrifugal Inline	0.09	40	0.0065	Roof

Notes

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Appendix B: Ventilation Rate Calculations

Density of Air (ρ) = 1.2 Kg/m3
Specific heat capacity of air (Cp) = 1.006 KJ/Kg.K
Zone Cooling Design supply air temperature = 14°C
Zone Heating Design supply air temperature = 40°C

Ground Floor

Space	ΔT	Cooling Load (KW)	Heating Load (KW)	Design Load (KW)	Mass Flow rate (Kg/s)	Volume Flow rate (m^3/s)
Lounge	8	1.6	0.9	1.6	0.198807157	0.165672631
Lobby	22	4.6	9	9	0.406651003	0.338875836
Closed office 1	8	4.9	2.7	4.9	0.608846918	0.507372432
Closed office 2	8	4.4	2.5	4.4	0.546719682	0.455599735
Meeting room 1	20	2.8	5.3	5.3	0.263419483	0.219516236
Meeting room 2	20	3	4.3	4.3	0.213717694	0.178098078

Floor 1

Space	ΔT	Cooling Load (KW)	Heating Load (KW)	Design Load (KW)	Mass Flow rate (Kg/s)	Volume Flow rate (m^3/s)
Open office	19	6.2	7	7	0.36622371	0.305186425
Closed office 1	8	1.3	1.3	1.3	0.161530815	0.134609013
Closed office 2	8	1.5	1	1.5	0.18638171	0.155318091
Closed office 3	8	3.6	2.1	3.6	0.447316103	0.372763419
Meeting room 1	20	2	3.8	3.8	0.188866799	0.157388999
Meeting room 2	20	2.3	4.4	4.4	0.218687873	0.182239894
Meeting room 3	20	1.6	3	3	0.149105368	0.124254473
Lobby	22	1.8	3.5	3.5	0.158142057	0.131785047

Floor 2

Space	ΔT	Cooling Load (KW)	Heating Load (KW)	Design Load (KW)	Mass Flow rate (Kg/s)	Volume Flow rate (m^3/s)
Open office	19	6.3	7.1	7.1	0.371455478	0.309546231
Closed office 1	20	1.3	1.4	1.4	0.069582505	0.057985421
Closed office 2	8	1.5	1	1.5	0.18638171	0.155318091
Closed office 3	8	3.6	2.2	3.6	0.447316103	0.372763419
Meeting room 1	20	2	3.8	3.8	0.188866799	0.157388999
Meeting room 2	20	2.3	4.5	4.5	0.223658052	0.18638171
Meeting room 3	20	1.5	2.9	2.9	0.144135189	0.120112657
Lobby	22	1.9	3.6	3.6	0.162660401	0.135550334

Appendix C: Duct Sizing Calculations

Ground Floor

Id	Item	Location	Volume flow rate (m^3/s)	ΔP (Pa/m)	Velocity (m/s)	Diameter (m)	Length (m)	Fitting Exit Velocity (m/s)	Fitting loss coefficient	Pressure loss (Pa)
1	Duct	Main trunk	1.88	0.45	5.2	0.66	1.4			0.63
2a	Tee-Straight	Main trunk						4.8	0.13	1.79712
2b	Tee-Branch	Main trunk						3.8	1.66	14.38224
3	Duct	Branch 1	0.46	0.45	3.8	0.39	3.12			1.404
4a	Tee-Straight	Branch 1						3.2	0.14	0.86016
4b	Tee-Branch	Branch 1						3.2	0.9	5.5296
5	Duct	Branch 1	0.23	0.45	3.2	0.3	1			0.45
6	Elbow	Branch 1						3.2	0.11	0.67584
7	Vertical duct	Branch 1	0.23	0.45	3.2	0.3	0.15			0.0675
8	Diffuser	Branch 1								25
9	Duct	Branch 1	0.23	0.45	3.2	0.3	2.17			0.9765
10	Elbow	Branch 1						3.2	0.11	0.67584
11	Vertical duct	Branch 1	0.23	0.45	3.2	0.3	0.15			0.0675
12	Diffuser	Branch 1								25
13	Duct	Main trunk	1.42	0.45	4.8	0.58	3.73			1.6785
14a	Tee-Straight	Main trunk						4.7	0.14	1.85556
14b	Tee-Branch	Main trunk						2.9	1.76	8.88096
15	Duct	Branch 2	0.18	0.45	2.9	0.27	7.42			3.339
16	Elbow	Branch 2						2.9	0.11	0.55506
17	Vertical duct	Branch 2	0.18	0.45	2.9	0.27	0.15			0.0675
18	Diffuser	Branch 2								25
19	Duct	Main trunk	1.24	0.45	4.7	0.56	1.61			0.7245
20a	Tee-Straight	Main trunk						4.6	0.15	1.9044
20b	Tee-Branch	Main trunk						2.8	1.59	7.47936
21	Duct	Branch 3	0.17	0.45	2.8	0.27	0.91			0.4095
22	Elbow	Branch 3						2.8	0.11	0.51744
23	Vertical duct	Branch 3	0.17	0.45	2.8	0.27	0.15			0.0675
24	Diffuser	Branch 3								25
25	Duct	Main trunk	1.07	0.45	4.6	0.53	1.62			0.729
26a	Tee-Straight	Main trunk						4	0.19	1.824
26b	Tee-Branch	Main trunk						3.9	0.99	9.03474
27	Duct	Branch 4	0.51	0.45	3.9	0.4	6.56			2.952
28a	Tee-Straight	Branch 4						3.2	0.15	0.9216
28b	Tee-Branch	Branch 4						3.2	1.08	6.63552
29	Duct	Branch 4	0.26	0.45	3.2	0.32	3.43			1.5435
30	Elbow	Branch 4						3.2	0.11	0.67584
31	Vertical duct	Branch 4	0.26	0.45	3.2	0.32	0.15			0.0675
32	Diffuser	Branch 4								25
33	Duct	Branch 4	0.26	0.45	3.2	0.32	0.97			0.4365
34	Elbow	Branch 4						3.2	0.11	0.67584
35	Vertical duct	Branch 4	0.26	0.45	3.2	0.32	0.15			0.0675
36	Diffuser	Branch 4		0.45	2.8	0.26	3.23			25
37	Duct	Main trunk	0.56	0.45	4	0.42	2.17			0.9765
38a	Tee-Straight	Main trunk						3.7	0.14	1.14996
38b	Tee-Branch	Main trunk						2.8	0.51	2.39904
39	Duct	Branch 5	0.17	0.45	2.8	0.27	1.76			0.792
40	Elbow	Branch 5			2.8	0.27		2.8	0.11	0.51744
41	Vertical duct	Branch 5	0.17	0.45	2.8	0.27	0.15			0.0675
42	Diffuser	Branch 5								25
43	Duct	Main trunk	0.39	0.45	3.7	0.37	1.83			0.8235
44	Elbow	Branch 6						3.7	0.11	0.90354
45	Duct	Branch 6	0.39	0.45	3.7	0.37	2.96			1.332
46a	Tee-Straight	Branch 6						3.2	0.14	0.86016
46b	Tee-Branch	Branch 6						2.8	1.04	4.89216
47	Duct	Branch 6	0.17	0.45	2.8	0.27	3.68			1.656
48	Elbow	Branch 6			2.8	0.27		2.8	0.11	0.51744
49	Vertical duct	Branch 6	0.17	0.45	2.8	0.27	0.15			0.0675
50	Diffuser	Branch 6			2.8	0.27				25
51	Duct	Branch 6	0.22	0.45	3.2	0.29	0.62			0.279
52	Elbow	Branch 6						3.2	0.11	0.67584
53	Vertical duct	Branch 6	0.22	0.45	3.2	0.29	0.15			0.0675
54	Diffuser	Branch 6								25

Frist and Second Floor

Id	Item	Location	Volume flow rate (m^3/s)	ΔP (Pa/m)	Velocity (m/s)	Diameter (m)	Length (m)	Fitting Exit Velocity (m/s)	Fitting loss coefficient	Pressure loss (Pa)
1	Duct	Main trunk	1.57	0.45	5	0.6	1.2			0.54
2a	Tee-Straight	Main trunk						4.7	0.13	1.72302
2b	Tee-Branch	Main trunk						3.5	1.97	14.4795
3	Duct	Branch 1	0.37	0.45	3.5	0.36	3.2			1.44
4a	Tee-Straight	Branch 1						2.9	0.14	0.70644
4b	Tee-Branch	Branch 1						2.9	0.84	4.23864
5	Duct	Branch 1	0.19	0.45	2.9	0.27	1			0.45
6	Elbow	Branch 1						2.9	0.11	0.55506
7	Vertical duct	Branch 1	0.19	0.45	2.9	0.27	0.2			0.09
8	Diffuser	Branch 1								25
9	Duct	Branch 1	0.18	0.45	2.9	0.27	2.15			0.9675
10	Elbow	Branch 1						2.9	0.11	0.55506
11	Duct	Branch 1	0.18	0.45	2.9	0.27	1.1			0.495
12	Elbow	Branch 1						2.9	0.11	0.55506
13	Vertical duct	Branch 1	0.18	0.45	2.9	0.27	0.2			0.09
14	Diffuser	Branch 1								25
15	Duct	Main trunk	1.2	0.45	4.7	0.55	1.34			0.603
16a	Tee-Straight	Main trunk						4.6	0.14	1.77744
16b	Tee-Branch	Main trunk						2.7	1.92	8.39808
17	Duct	Branch 2	0.13	0.45	2.7	0.24	0.92			0.414
18	Elbow	Branch 2						2.7	0.11	0.48114
19	Vertical duct	Branch 2	0.13	0.45	2.7	0.24	0.2			0.09

20	Diffuser	Branch 2								25
21	Duct	Main trunk	1.07	0.45	4.6	0.54	1.26			0.567
22a	Tee-Straight	Main trunk						4.4	0.14	1.62624
22b	Tee-Branch	Main trunk						3.3	1.02	6.66468
23	Duct	Branch 3	0.28	0.45	3.3	0.33	3.25			1.4625
24a	Tee-Straight	Branch 3						2.8	0.14	0.65856
24b	Tee-Branch	Branch 3						2.7	1.06	4.63644
25	Duct	Branch 3	0.12	0.45	2.7	0.24	0.67			0.3015
26	Elbow	Branch 3						2.7	0.11	0.48114
27	Vertical duct	Branch 3	0.12	0.45	2.7	0.24	0.2			0.09
28	Diffuser	Branch 3								25
29	Duct	Branch 3	0.16	0.45	2.8	0.26	3.77			1.6965
30	Elbow	Branch 3						2.8	0.11	0.51744
31	Duct	Branch 3	0.16	0.45	2.8	0.26	0.64			0.288
32	Elbow	Branch 3						2.8	0.11	0.51744
33	Vertical duct	Branch 3	0.16	0.45	2.8	0.26	0.2			0.09
34	Diffuser	Branch 3								25
35	Duct	Main trunk	0.79	0.45	4.4	0.48	1.65			0.7425
36a	Tee-Straight	Main trunk						4.2	0.13	1.37592
36b	Tee-Branch	Main trunk						2.9	1.5	7.569
37	Duct	Branch 4	0.18	0.45	2.9	0.27	3.44			1.548
38	Elbow	Branch 4						2.9	0.11	0.55506
39	Vertical duct	Branch 4	0.18	0.45	2.9	0.27	0.2			0.09
40	Diffuser	Branch 4								25
41	Duct	Main trunk	0.61	0.45	4.2	0.44	2			0.9
42a	Tee-Straight	Main trunk						3.8	0.14	1.21296
42b	Tee-Branch	Main trunk						2.8	0.96	4.51584
43	Duct	Branch 5	0.16	0.45	2.8	0.26	3.23			1.4535
44	Elbow	Branch 5						2.8	0.11	0.51744
45	Duct	Branch 5	0.16	0.45	2.8	0.26	0.79			0.3555
46	Elbow	Branch 5						2.8	0.11	0.51744
47	Vertical duct	Branch 5	0.16	0.45	2.8	0.26	0.2			0.09
48	Diffuser	Branch 5								25
49	Duct	Main trunk	0.45	0.45	3.8	0.38	1.76			0.792
50a	Tee-Straight	Main trunk						2.7	0.15	0.6561
50b	Tee-Branch	Main trunk						3.3	0.74	4.83516
51	Duct	Branch 6	0.32	0.45	3.3	0.33	2			0.9
52a	Tee-Straight	Branch 6						2.8	0.14	0.65856
52b	Tee-Branch	Branch 6						2.8	1.04	4.89216
53	Duct	Branch 6	0.16	0.45	2.8	0.26	1.05			0.4725
54	Elbow	Branch 6						2.8	0.11	0.51744
55	Vertical duct	Branch 6	0.16	0.45	2.8	0.26	0.2			0.09
56	Diffuser	Branch 6								25
57	Duct	Branch 6	0.16	0.45	2.8	0.26	4.28			1.926
58	Elbow	Branch 6						2.8	0.11	0.51744
59	Vertical duct	Branch 6	0.16	0.45	2.8	0.26	0.2			0.09
60	Diffuser	Branch 6								25
61	Duct	Main trunk	0.13	0.45	2.7	0.24	1.95			0.8775
62	Elbow	Branch 7						2.7	0.11	0.48114
63	Duct	Branch 7	0.13	0.45	2.7	0.24	3.2			1.44
64	Elbow	Branch 7						2.7	0.11	0.48114
65	Vertical duct	Branch 7	0.13	0.45	2.7	0.24	0.2			0.09
66	Diffuser	Branch 7								25

Supply and Return risers

Id	Item	ne flow rate (	ΔP (Pa/m)	elocity (m/s	Diameter (m)	Length (m)	Exit Velocit	Fitting loss coefficient	Pressure loss (Pa)
1	Duct	5.02	0.45	6.4	0.95	0.45			0.2025
2	Tee-Straight						6	0.13	2.808
3	Tee-Branch						5	1.12	16.8
4	Duct	3.45	0.45	6	0.82	3.17			1.4265
5	Tee-Straight						5.2	0.14	2.27136
6	Tee-Branch						5	0.97	14.55
7	Duct	1.88	0.45	5.2	0.66	3.03			1.3635
8	Elbow						5.2	0.11	1.78464

Appendix D: Pressure Loss Analysis

Pressure drop per Volume Control Damper(VCD) = 10Pa

Ground Floor

Terminal diffuser	Ductwork Pressure loss (Pa)	Number of VCDs on route	VCD Pressure loss (Pa)	Total pressure loss (Pa)
8	48.13918	3	30	78.13918
12	43.99624	3	30	73.99624
18	41.94814	2	20	61.94814
24	40.15948	2	20	60.15948
32	55.22818	3	30	85.22818
36	48.40726	3	30	78.40726
42	40.89556	2	20	60.89556
50	48.46168	3	30	78.46168
54	43.21108	3	30	73.21108

Floor Index run pressure loss = 85.2Pa

First and Second Floor

Terminal diffuser	Ductwork Pressure loss (Pa)	Number of VCDs on route	VCD Pressure loss (Pa)	Total Pressure loss (Pa)
14	44.82856	3	30	74.82856
8	46.7932	3	30	76.7932
20	37.24924	2	20	57.24924
34	42.10558	3	30	72.10558
28	43.84672	3	30	73.84672
40	42.34126	2	20	62.34126
48	42.30484	2	20	62.30484
56	48.56734	3	30	78.56734
60	45.78724	3	30	75.78724
66	40.88596	2	20	60.88596

Floor Index run pressure loss = 7806Pa

Pressure drop from fan to each floor's index run

Floor	Pressure loss (Pa)
Floor 2	95.56984
Floor 1	97.55434
Ground floor	95.08468

Appendix E: Return Grille Sizing

Ground Floor

Space	Volume flow rate (m^3/s)	Target velocity (m/s)	Required face area (m^2)	Total area (m2)	Chosen dimensions (mm)
Lounge	0.17	2	0.085	0.121428571	350x350
Lobby	0.34	2	0.17	0.242857143	500x500
Closed office 1	0.51	2	0.255	0.364285714	600x600
Closed office 2	0.46	2	0.23	0.328571429	600x600
Meeting room 1	0.22	2	0.11	0.157142857	400x400
Meeting room 2	0.18	2	0.09	0.128571429	400x400

First and Second Floor

Space	Volume flow rate (m^3/s)	Target velocity (m/s)	Required face area (m^2)	Total area (m2)	Chosen dimensions (mm)
Open office	0.32	2	0.16	0.228571429	500x500
Closed office 1	0.13	2	0.065	0.092857143	350x350
Closed office 2	0.16	2	0.08	0.114285714	350x350
Closed office 3	0.37	2	0.185	0.264285714	550x550
Meeting room 1	0.16	2	0.08	0.114285714	350x350
Meeting room 2	0.18	2	0.09	0.128571429	400x400
Meeting room 3	0.12	2	0.06	0.085714286	300x300
Lobby	0.13	2	0.065	0.092857143	350x350